

Exploratory Data Analysis (EDA) Report for the Adult Income Prediction Dataset

1.0 Introduction

This report presents an exploratory data analysis of the Adult Income Prediction dataset, sourced from Kaggle. The dataset contains demographic and employment-related information about individuals from the 1994 U.S. Census. The primary goal of this analysis is to understand the factors that influence an individual's income level, categorized as either $\leq 50K$ (less than or equal to \$50,000) or $> 50K$ (greater than \$50,000). This EDA will uncover patterns and provide insights that can inform policy formulation processes and also be instrumental in building predictive models for income classification.

The EDA was performed using the Python programming language and Python libraries such as pandas, matplotlib and seaborn.

2.0 About the Dataset

The dataset consists of 32,561 records and 15 features, a mix of categorical and numerical data types. The features are as follows:

Attribute	Description
age	The age of the individual.
workclass	The type of employment (e.g., Private, Self-emp-not-inc, etc.).
fnlwgt	"Final weight", representing the number of people the census believes the entry represents.
education	The highest level of education achieved (e.g., Bachelors, HS-grad, etc.).
education-num	The numerical representation of the education level.
marital-status	The marital status of the individual.
occupation	The individual's occupation (e.g., Tech-support, Craft-repair, etc.).
relationship	The individual's relationship in their household (e.g., Wife, Own-child, etc.).
race	The individual's race (e.g., White, Asian-Pac-Islander, etc.).
sex	The individual's gender (Male, Female).
capital-gain	The individual's capital gains.
capital-loss	The individual's capital losses.
hours-per-week	The number of hours the individual works per week.
native-country	The individual's native country.
income	The target variable, indicating income level ($\leq 50K$ or $> 50K$).

3.0 Data Understanding and Preparation

An initial look-through of the dataset reveals the following:

- **Data Types:** The data is a mixture of numerical and categorical features. The categorical features will need to be encoded if machine learning models are to be developed.

- **Missing Values:** The dataset contains missing values denoted by a “?” in the *workclass*, *occupation*, and *native-country* columns. Since these values cannot be assumed, the records containing these missing values were removed to improve the quality of the analysis.
- **Data Cleaning:** After the cleaning process, 92.63% of the initial data remained for the analysis, which was a satisfactory value.
- **Feature correlation:** The correlation of the numerical columns was calculated and visualized using a heatmap. The heatmap revealed a weak positive relationship between age and hours worked per week, and also, education level and hours worked per week had a weak positive relationship. In addition, capital gain and capital loss both had a weak positive relationship with education level, respectively.

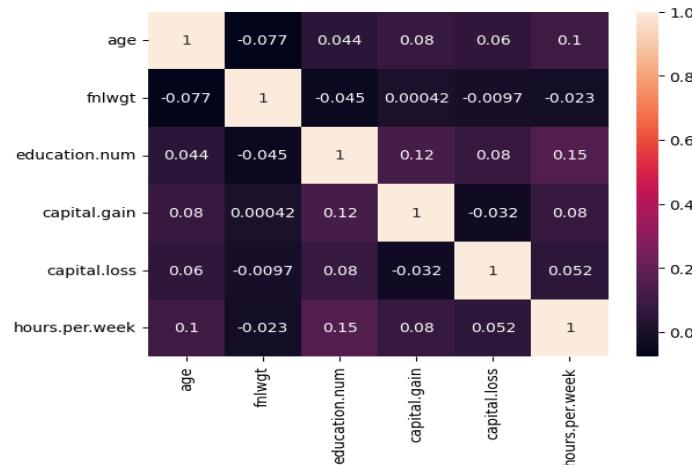


Figure 1. The correlation heatmap of the numerical features

4.0 Answering Important Questions

Some key questions were asked to understand the relationships between different features and the income earned per year.

4.1 How does hours worked per week influence the likelihood of earning over or less than \$50K per year?

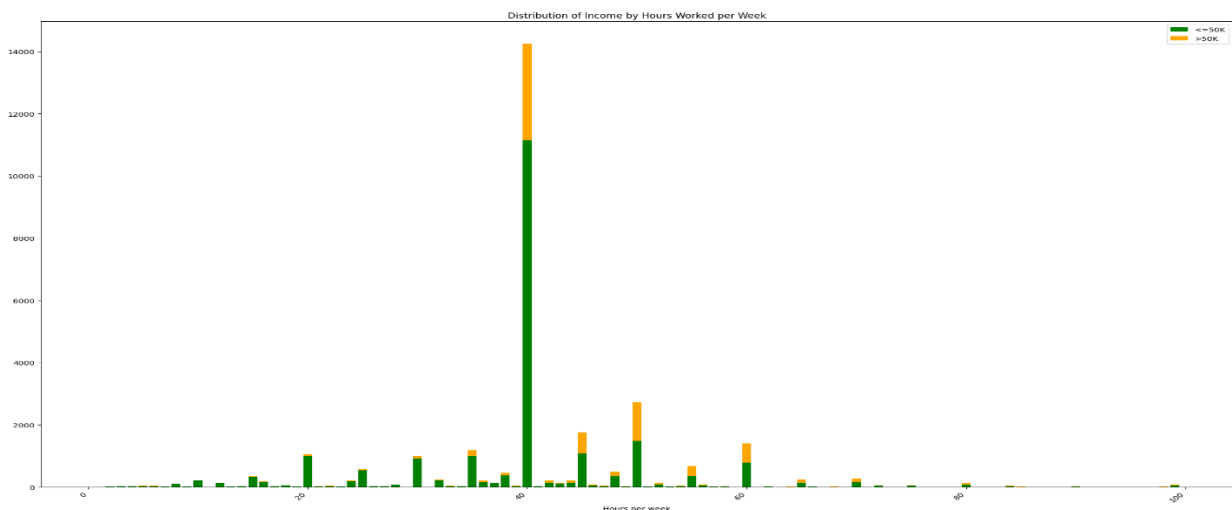


Figure 2. Distribution of income and work hours per week

There is a prevalence of low-income earners across the weekly working hours of the population, as shown in Figure 2. The standard 40-hour work week recorded the most individuals with low ($\leq \$50K$) and high ($> \$50K$) income earners. Also, the majority of individuals worked between 20 and 60 hours per week. This specific analysis shows that working more hours per week doesn't always translate to earning more money.

4.2 How does education level impact the likelihood of earning over or less than \$50K per year?

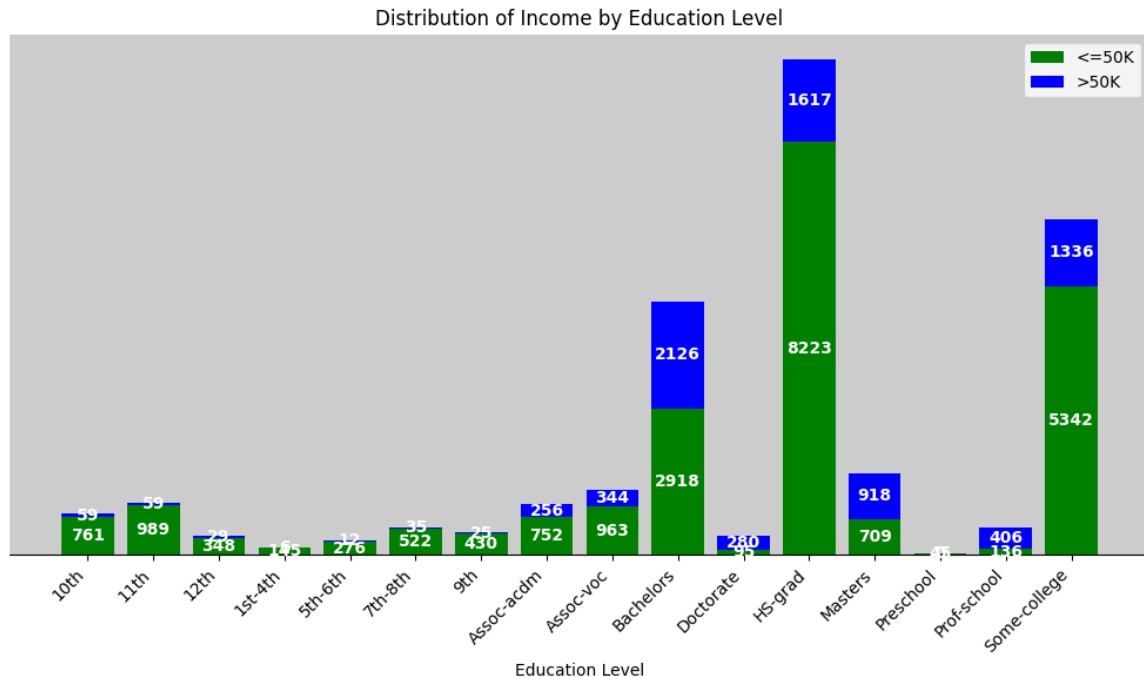


Figure 3. Distribution of income and education level

Education level is one of the strongest predictors of income in this dataset. Higher levels of educational attainment are directly linked with a much higher probability of earning over \$50K. Bachelor's degree holders had 2,126 persons (representing 42% of their group) in the high-income bracket, followed by high school graduates at 1,617 persons (16% of their group) and college dropouts (some-college) at 1,336 persons (20% of their group). By contrast, the high school graduate group had the highest number of low-income earners with 8,223 persons (84% of their group), followed by college dropouts at 5,342 persons (80% of their group) and bachelor's degree holders at 2,918 persons (58% of their group). Only the doctorate, master's, and professional school groups had the larger percentage of their population in the high-income bracket, indicating that these particular education levels are more lucrative than others.

4.3 Is there a significant income gap between genders?



Figure 4. Distribution of income and gender

The dataset reveals a significant income disparity between males and females. 31% of the men in the dataset earn more than \$50K, while 11% of the women earn more than \$50K. Consequently, males in the high-income bracket are approximately 6 times more than women in the same income bracket. This substantial gap indicates a big difference in earning potential between the two genders as captured in the 1994 census data.

4.4 Which occupations have the highest earning potential?

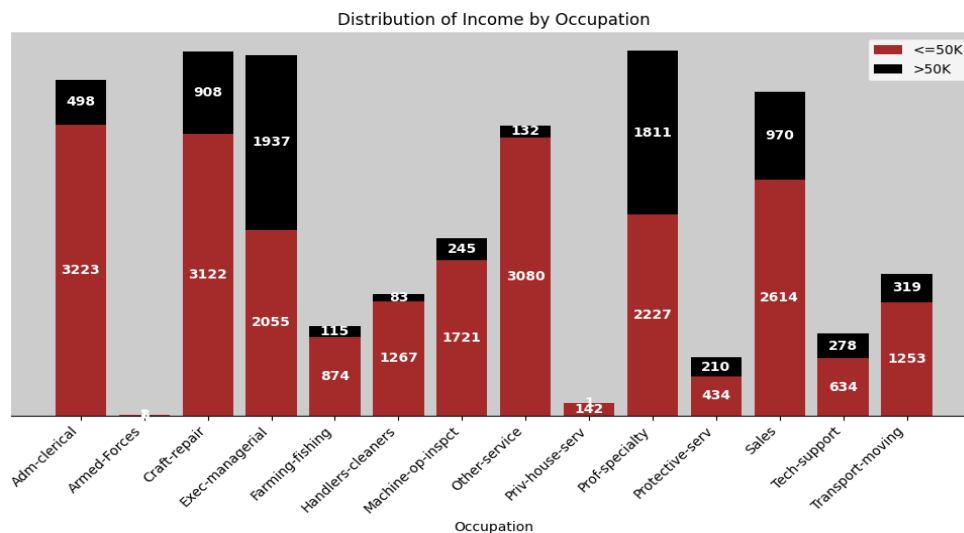


Figure 5. Distribution of income and occupation

The data reveals that the likelihood of earning over \$50K varies greatly by occupation. Executive managerial (1,937 people at 48% of their group) and specialized professional (1,811 people at 45% of their group) roles show the highest earning potential.

In contrast, occupations like Priv-house-serv (Private house service), Other-service, and Handlers-cleaners have the lowest earning potential, with less than 7% of individuals in those roles earning over \$50K. Also, only 1% of armed forces personnel captured in the dataset are in the high-income bracket. This piece of information indicates that people with seemingly low-grade jobs earn way less compared with people with high-grade jobs.

4.5 How does marital status and relationship role influence income?

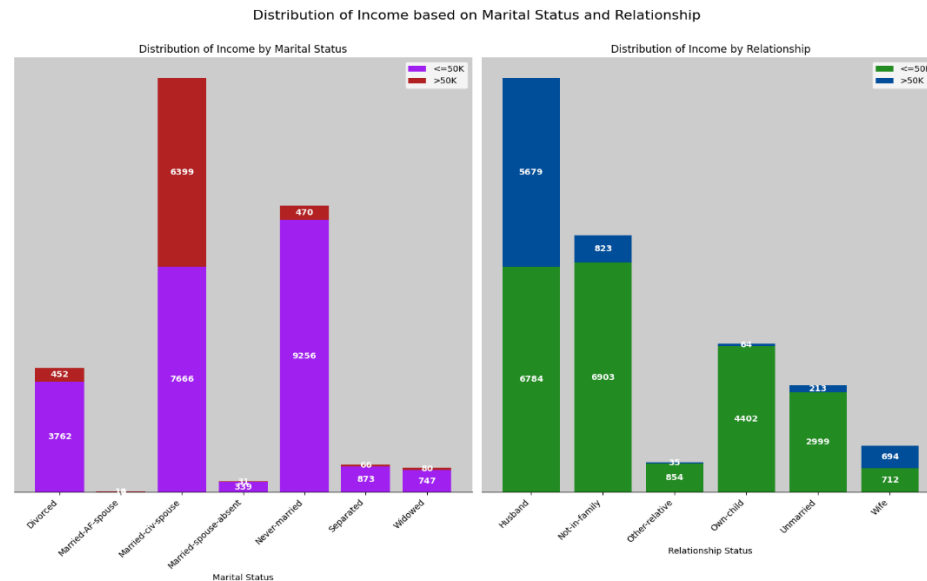


Figure 6. Distribution of income and marital status and relationship status

Marital status also appears to be a strong indicator of income level. Individuals who are married and living with a civilian spouse have the highest number of high-income earners (45% of their group). Other relatively high-earning marital groups are the divorced (11% of their group) and never-married (5% of their group) groups. The relationship status reinforces this finding. The *husband* role has the most individuals earning over \$50k, representing 46% of the group, while the *wife* role is the second highest, representing 49% of its group. These findings are supported by the emergence of the *Married-civ-spouse* marital status as having the highest income earners. This bit of analysis shows that married households, as represented by individuals in these roles, tend to have higher earnings compared to single, divorced, or widowed individuals.

4.6 Are there income disparities across different racial groups?

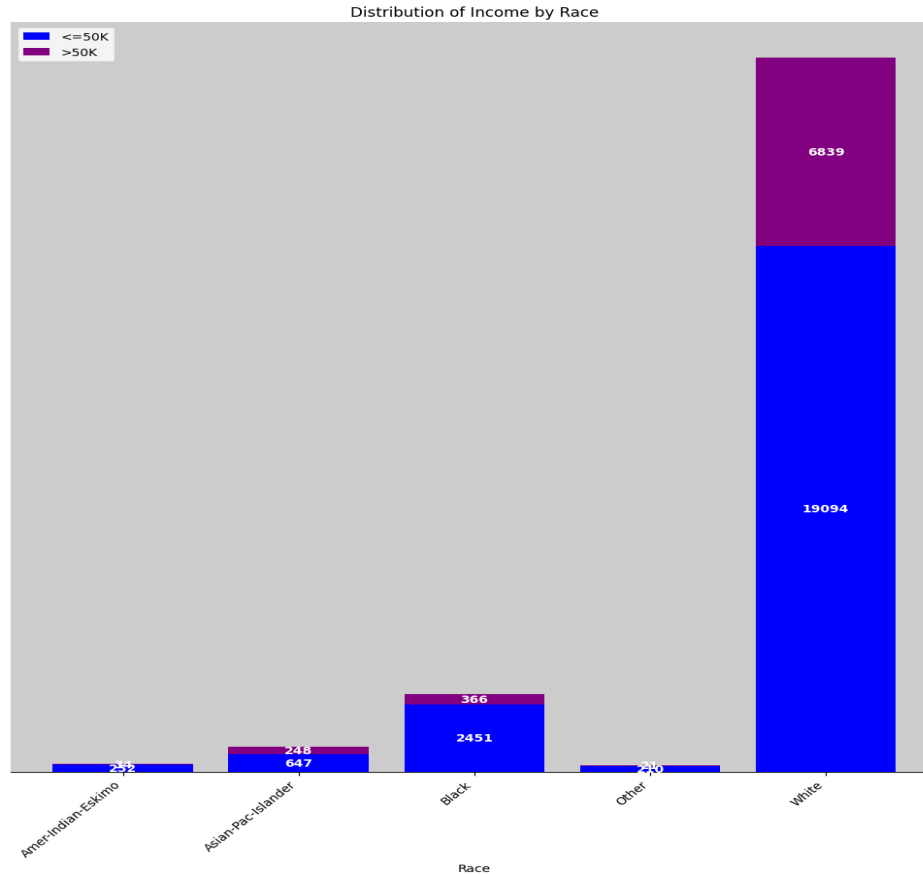


Figure 7. Distribution of income and race

The data indicate clear income differences across the racial groups represented in the dataset. Individuals identified as *White* (having 6,839 persons) and *Asian-Pac-Islander* (having 248 persons) have a proportionally higher representation in the >\$50K income category (26% and 28% respectively) compared to individuals from other racial backgrounds. By contrast, individuals identified as *Other* race had the highest proportion of persons in the low-income bracket at 91%. Other races identified with having significant proportions of their group in the low-income bracket are *Amer-Indian-Eskimo* and *Black* at 88% and 87% respectively. This analysis confirms the fact that the US is a white-dominated country and therefore a higher number of whites should have high-paying jobs, and conversely, significantly lower-paying jobs.

5.0 Conclusion

The exploratory data analysis of the Adult Income Prediction dataset revealed several key factors that are strong indicators of an individual's income level based on the 1994 US census data. The most influential factors appear to be education level, occupation, and marital status.

Surprisingly, hours worked per week didn't have a significant effect on income level, except for the high volume of people on the standard 40-hour work week regimen. Also, demographic factors such as gender and race show a significant correlation with income, highlighting potential social and economic disparities.

Therefore, the data gives an image of the characteristic high-income earner in the US in 1994: a married individual, typically male and identified as White or Asian-Pacific Islander, with an advanced degree (Bachelor's, Master's, or Doctorate), and working in an executive, managerial, or specialized professional role for the standard 40-hour work week.

6.0 Recommendations

Based on the findings of this EDA, the following recommendations are presented.

- There is a need for targeted governmental interventions to address the significant income differences between racial groups (e.g., White and Asian-Pac-Islander groups earning proportionally more than Black and Amer-Indian-Eskimo groups).
- Governments should foster equity through robust enforcement of anti-discrimination laws in hiring and promotion.
- Equal pay legislation should be empowered and enforced to challenge pay discrimination between the male and female genders.
- For further development of machine learning models for the dataset, numerical columns like age and hours-per-week should be standardized to ensure high accuracy of the developed models.